

## 20.2 Software tools for creation and editing of animations

Software tools can be used to create animations for a range of applications, or purposes, and audiences. An animation's purpose and audience should be carefully considered to ensure their requirements and needs are met. For example, an animation to show safety information to adults before an aeroplane flight should clearly display all information without distraction and the viewer should be made aware of what they need to do in an emergency. For an animation to advertise a children's television programme, the characters within the programme might be included and quickly move around the screen, including using bright colours and engaging sounds.

### Configuring the stage for an animation

The area in animation software where you draw and create an animation is called the **stage**.

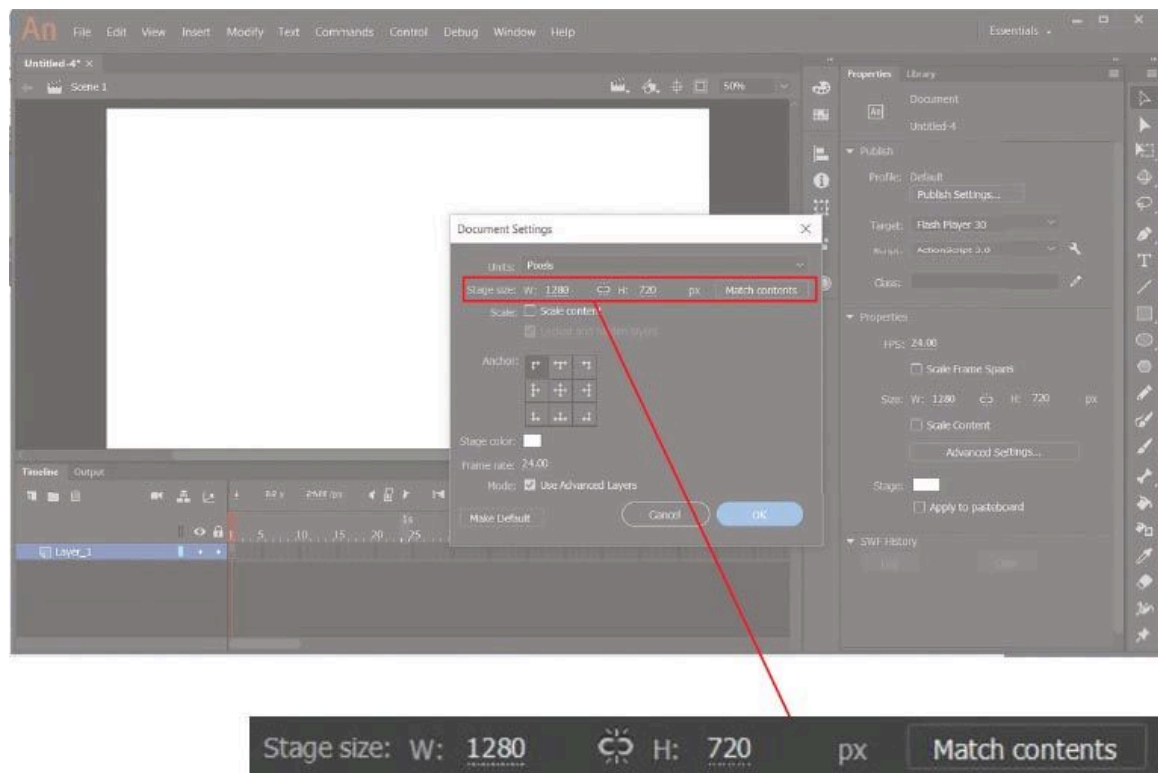
When Adobe Animate is opened, the stage is shown as a white rectangle. Any objects that should appear as part of an animation must be in this stage.

#### Setting the size of the stage

The size of the stage is measured in pixels. The size can be changed by increasing or decreasing width and/or height, which is usually measured in pixels.

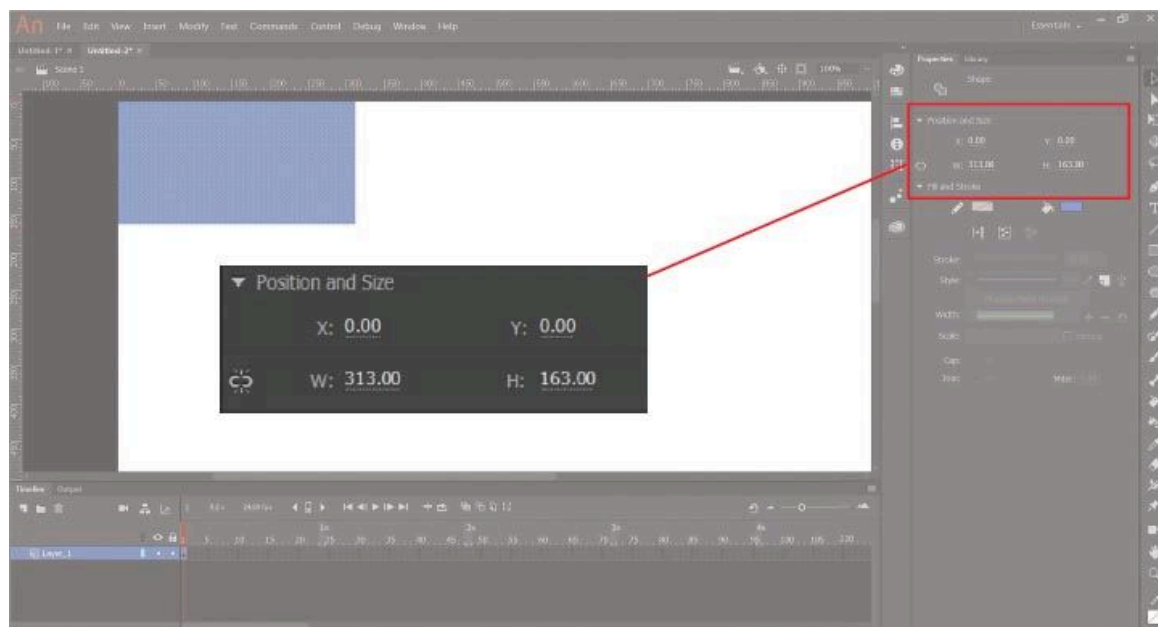
To change the stage size in Adobe Animate, click **Modify** and then **Document**.

The width and height can be changed in stage size. The drop-down menu changes the units from pixels to centimetres or inches if required (see Figure 20.7).



**Figure 20.7:** Changing size of the stage.

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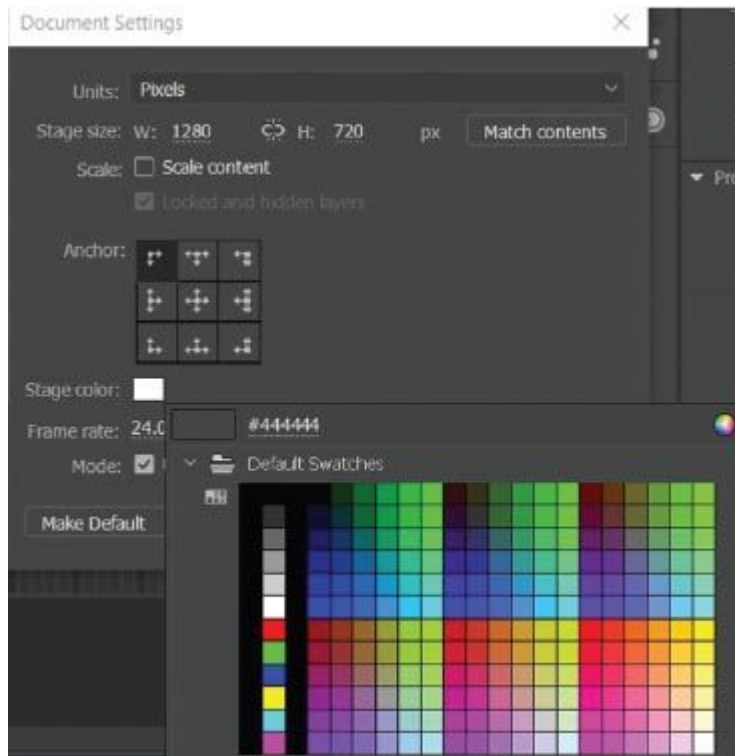
**Figure 20.8:** Position and size options.

The stage has a set of *x*- and *y*-**coordinates** that can be used to position objects with more accuracy.

The top-left corner of the stage has the coordinates (0, 0) and these then increase as you move down and across the image. In Figure 20.8, the ruler is visible at the top and left of the screen; this gives you the coordinates. This blue rectangle in Figure 20.8 starts at *x*-position 0 (the left-hand edge of the stage) and its width is 313, which shows how far to the right the rectangle goes. The blue rectangle also starts at *y*-position 0 (the top edge of the stage) and its height goes down to 163.

## Setting the colour of the stage

When **Modify** and then **Document** are selected, the colour of the stage can be changed by clicking the **Stage colour** box. This will bring up a range of colours options (see Figure 20.9). Click the sphere in the top right to access more colours, or type the colour's hexadecimal code directly into the box.



**Figure 20.9:** Colour options.

## Setting the aspect ratio

The aspect ratio relates to the relationship between the width and height of the stage. There are standard ratios depending on what the animation is used for. For example, if producing a film for widescreen televisions, an appropriate aspect ratio needs to be selected so the animation does not appear incorrectly. It is important to consider where your animation is most likely to be viewed when setting up the stage, which will enable you to decide on a suitable aspect ratio. When starting a project in Adobe Animate some common options are displayed to choose from (see Figure 20.10).

Common aspect ratios are shown in Table 20.2.

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| Ratio  | Description                                                                                                                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4:3    | This is the aspect ratio used in older televisions, so is rarely used now. It is 4 units wide to 3 units high. The stage could be 2000 pixels wide by 1500 pixels high.                                              |
| 16:9   | This is the ratio that is currently most widely used for TV. It is also commonly used in computer monitors and laptops. It is 16 units wide to 9 units high. The stage could be 1600 pixels wide by 900 pixels high. |
| 2.35:1 | This is widescreen and used for films.                                                                                                                                                                               |

|       |                                                                                          |
|-------|------------------------------------------------------------------------------------------|
|       | It is 2.35 units wide to 1 unit high.                                                    |
| 16:10 | This is commonly used for smart phone displays.<br>It is 16 units wide to 10 units high. |

Table 20.2: Common aspect ratios.

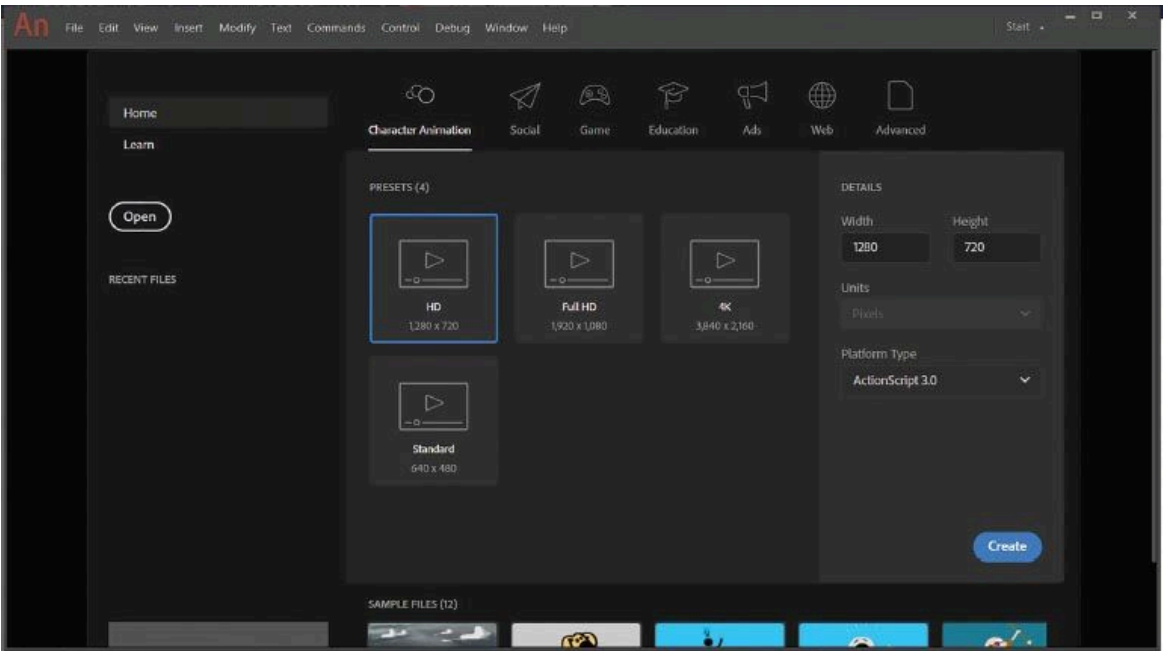


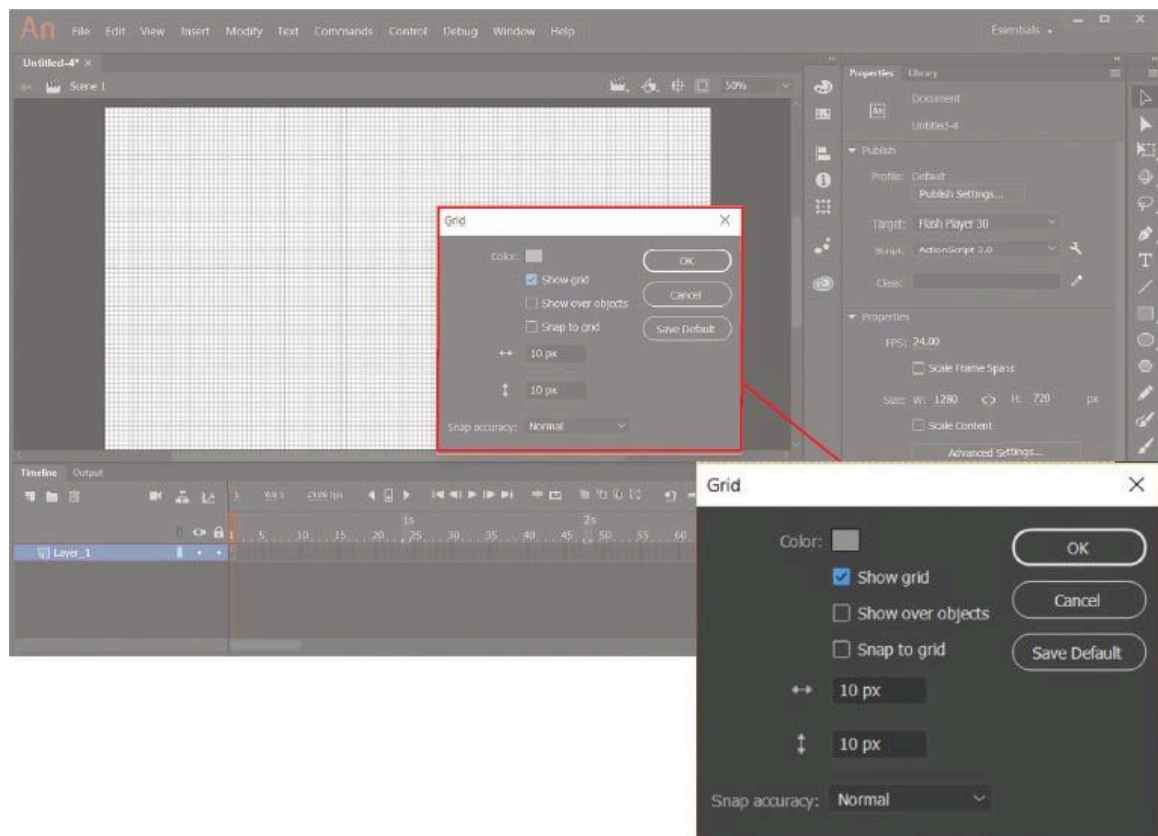
Figure 20.10: Common aspect ratio options when starting a project.

Using rulers, guides, grid settings and snapping options

The stage has several features that can help lay out animations.

**Rulers** enable the size and position of items to be viewed. Rulers can be turned on and off by clicking **View** from the top menu, then clicking **Rulers**.

The **grid** option displays a grid on the background of the stage. This can help align objects to the same gridline. To turn the grid on or off, click **View** from the top menu, then **Grid** and then **Show Grid**. The grid can be altered, such as by changing the size of the grid's squares or how it is displayed, by selecting **Edit Grid**, which brings up a menu. In Figure 20.11, there is 1 grid line for every 10 pixels, although this can be increased or decreased.



**Figure 20.11:** Grid options.

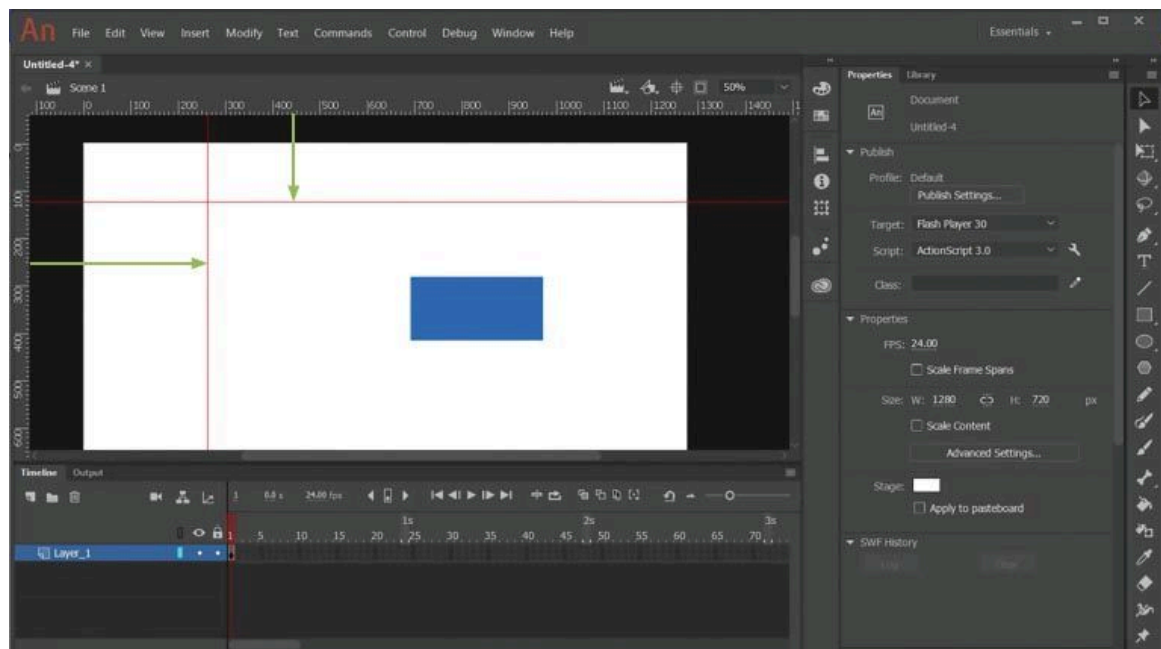
The **snapping** option, [Snap to grid](#), makes sure all objects align to one of the grid lines by slightly adjusting their position if required.

Other snapping options can be accessed from the [View](#) menu. These allow Adobe Animate to align objects automatically to:

- display lines from the object to let you line it up with other elements, using the option [Snap Align](#)
- other objects, using [Snap to Objects](#)
- specific pixels or lines of pixels, using [Snap to Pixels](#).

**Guides** can also be added to the stage. Guides are lines that can be positioned to help you line up your objects. Turn on guides by selecting [View](#) from the top menu, then select [Guides](#) and then [Show Guides](#).

The lines are then dragged down from the top of the screen, as shown by the green arrows in Figure 20.12. The red lines are the guides that then appear. You can change the colour of the guide lines by choosing [Edit Guides](#) from the [Guides](#) menu.



**Figure 20.12:** Positioning guides.

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## Questions

- 7 What is a frame in animation?
- 8 Describe what a ratio of 4:3 means and when is it used.
- 9 Name two other common ratios and when they are used.
- 10 Explain the purpose of rulers, grids and guides.
- 11 Explain the advantage(s) of using snapping.

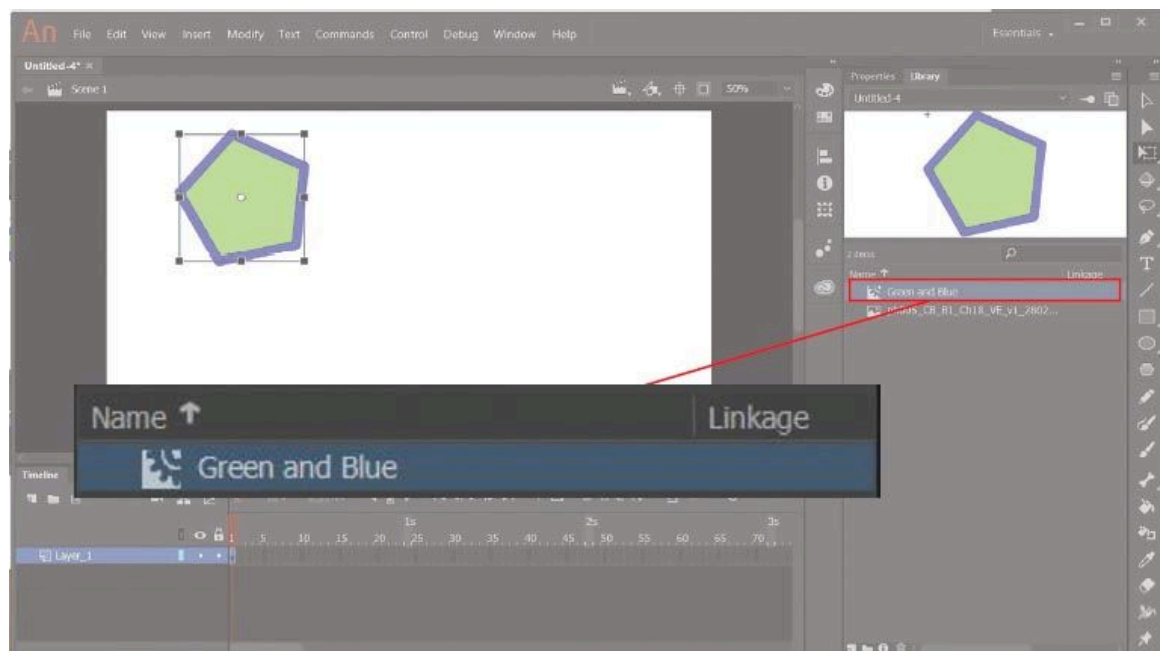
## Importing and creating vector objects

To move and manipulate objects within an animation, each object is named, which is known as a **symbol**. Figure 20.13 shows how to name a symbol. If required, multiple objects can be grouped together into one symbol. To convert a selection to a symbol, use the selection tool, then right click the item. Choose **Convert to Symbol**, then give the symbol a suitable name.

All symbols appear in the **library**, which is a record of all the symbols in the animation. New instances of a symbol can be created by dragging its name onto the stage. Up to 20 instances of the same symbol can be created and animated separately, although if one symbol is altered, the rest will also change.

Vector graphics are based on mathematical equations to determine the image's points, lines and curves, rather than using individual pixels, which means they can be resized without pixelating or reducing the quality of the image. There is more information about vector graphics in [Chapter 19](#), Graphics creation.





**Figure 20.13:** Naming a symbol.

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## Tracing bitmaps

Existing images, such as a photograph, can be imported into the library, or onto the stage. To import an image, click **File** from the top menu bar, then **Import** and either **Import to Stage** or **Import to Library**. Find the image and click **Open** to import it.

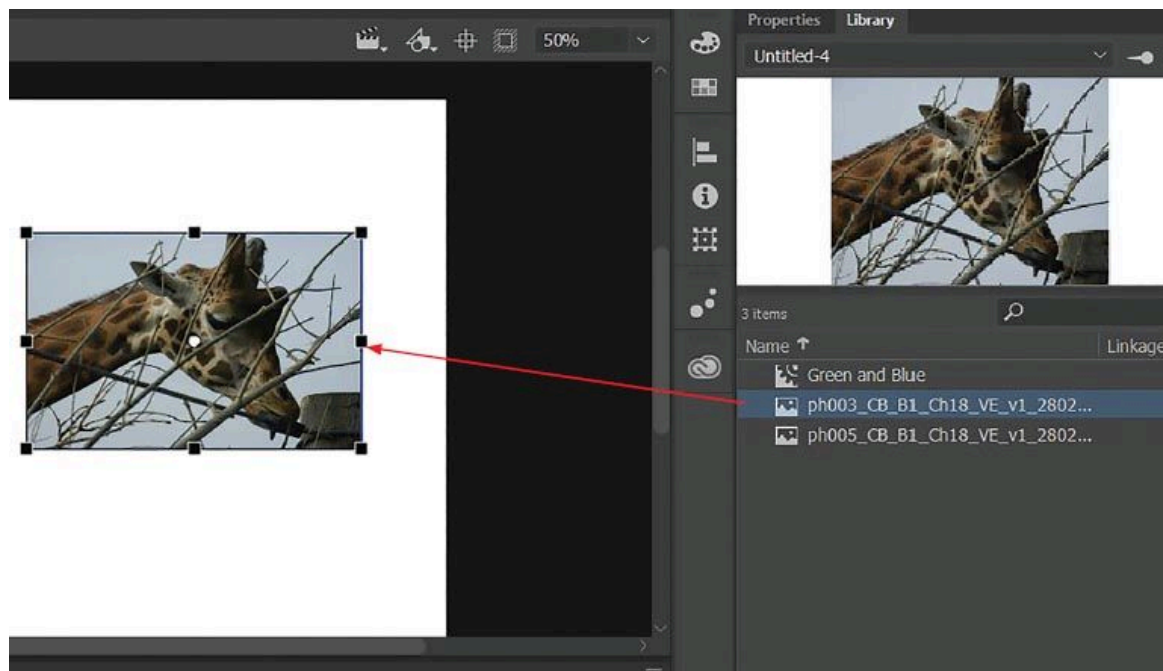
Bitmaps cannot be animated like vector graphics, as they are not made of coordinates and calculations. They might also pixelate when enlarged. To try to prevent this, bitmap images can be converted into a vector graphic. This is called **tracing bitmaps**.

A disadvantage of exporting to a vector graphic is that some of the detail is lost. For example, if the image is a photograph, then it might not appear exactly as the original. It will also not be possible to edit the image in the same way, such as by using the bitmap image manipulation tools.

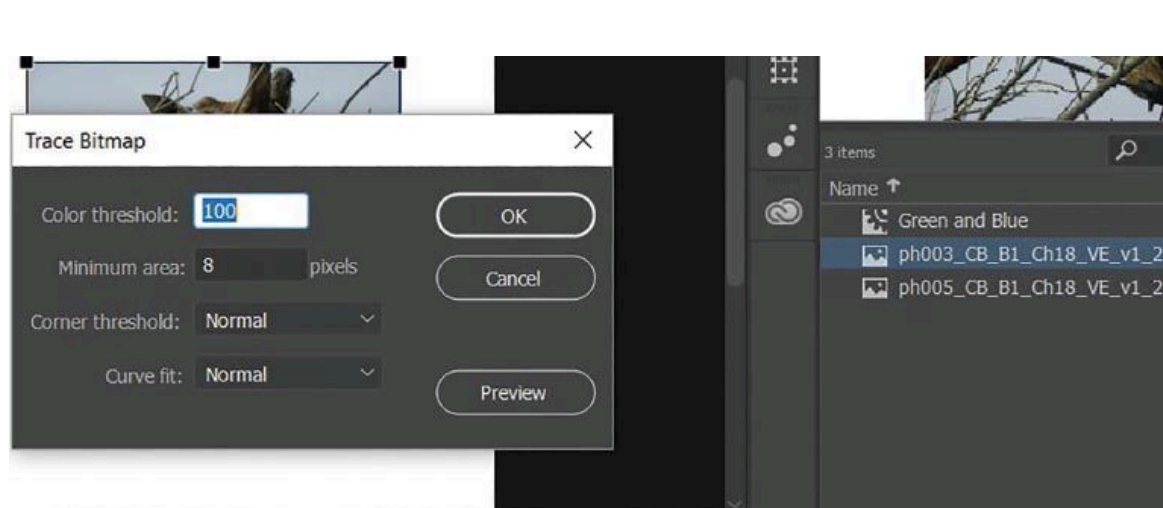
To trace a bitmap, first drag the bitmap image from the library onto the stage, as shown in Figure 20.14.

With the object selected, select **Modify** from the top menu, and then **Bitmap**, then **Trace Bitmap**.

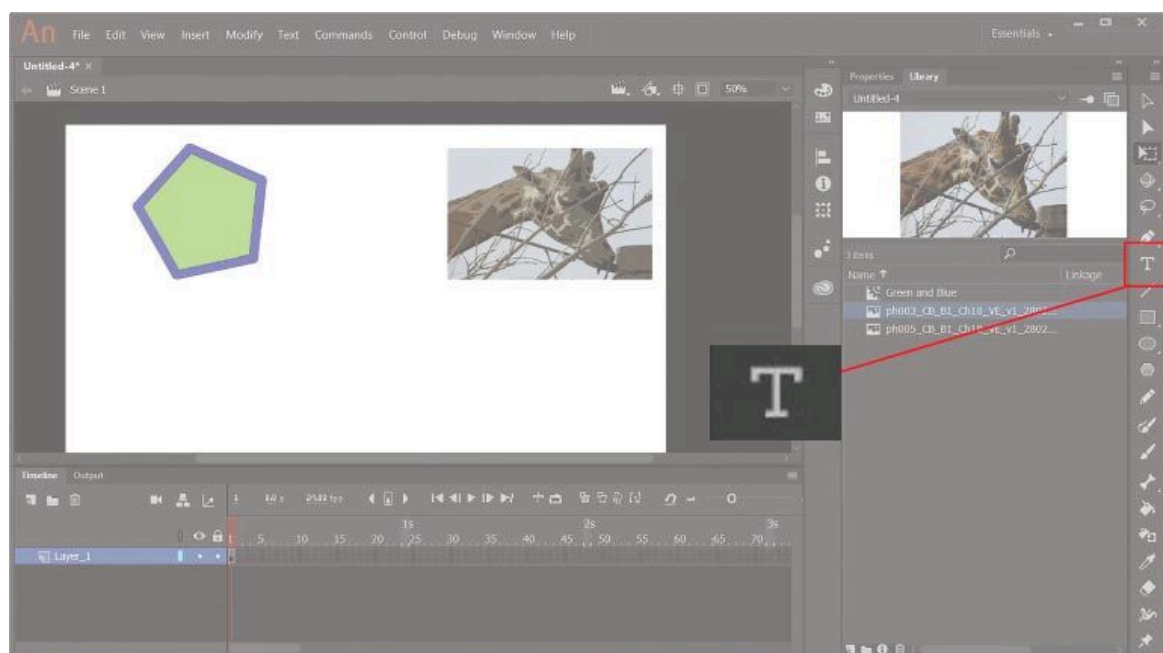
Within the options shown on screen, as shown in Figure 20.15, the lower the colour threshold, the more accurate it will be. Adobe Animate will usually give the most appropriate threshold automatically. Click **OK** and then the bitmap will be turned into a vector. Sections of the image can be selected and manipulated independently.



**Figure 20.14:** Getting an image from the library.



**Figure 20.15:** Bitmap manipulation options.





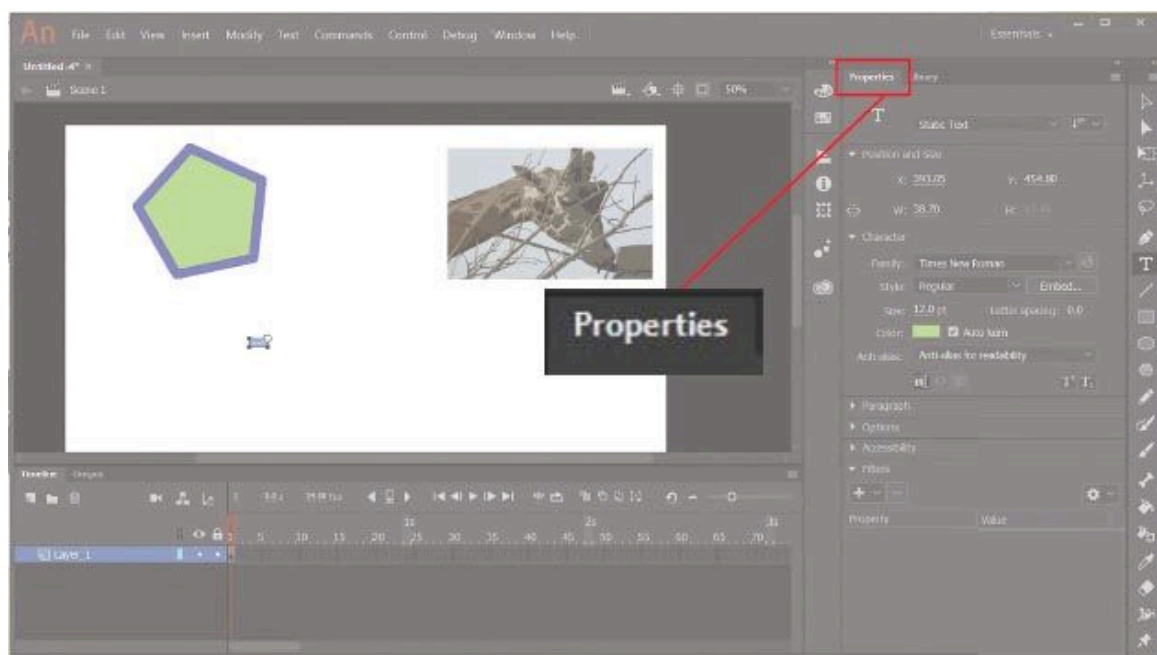
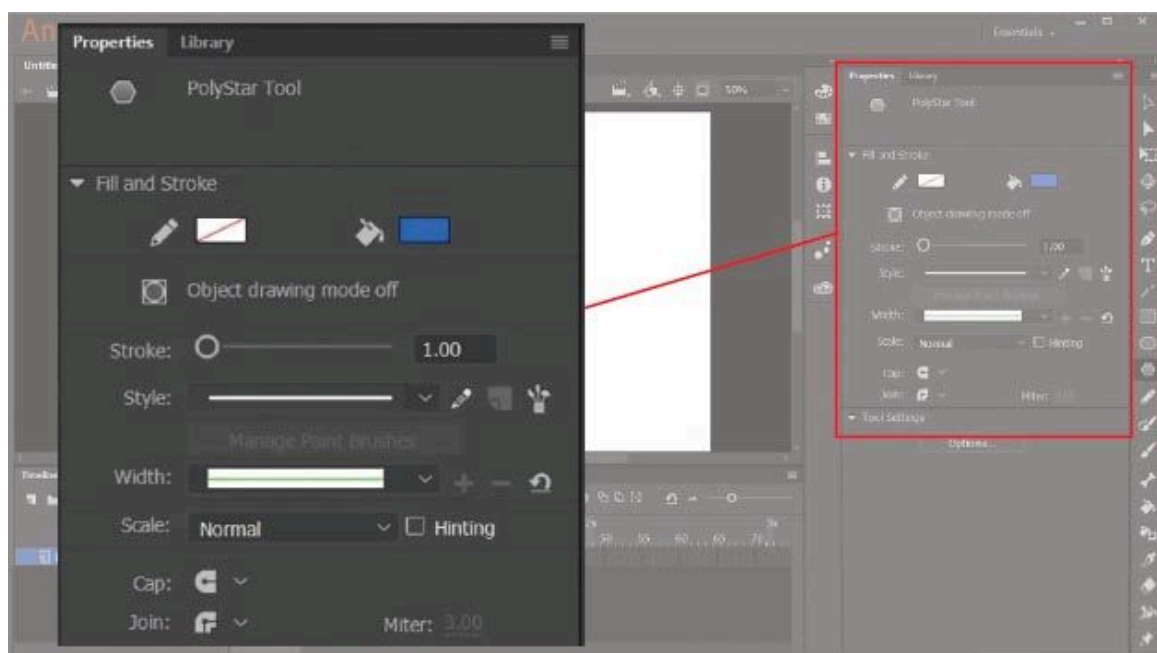
**Figure 20.16:** Text tool button.

## Adding text

Text can be added to the stage using the text tool, which lets you edit the style, size and colour. Text is treated in the same way as objects, so you can change the opacity (lack of transparency) and manipulate the object in the same way as vector images.

To add text, click the **T** on the toolbar (see Figure 20.16), then click on the stage where you want the text to appear.

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**Figure 20.17:** Text properties options.**Figure 20.18:** Properties options.

Click **Properties** to change properties, such as the font name, size and colour (see Figure 20.17).

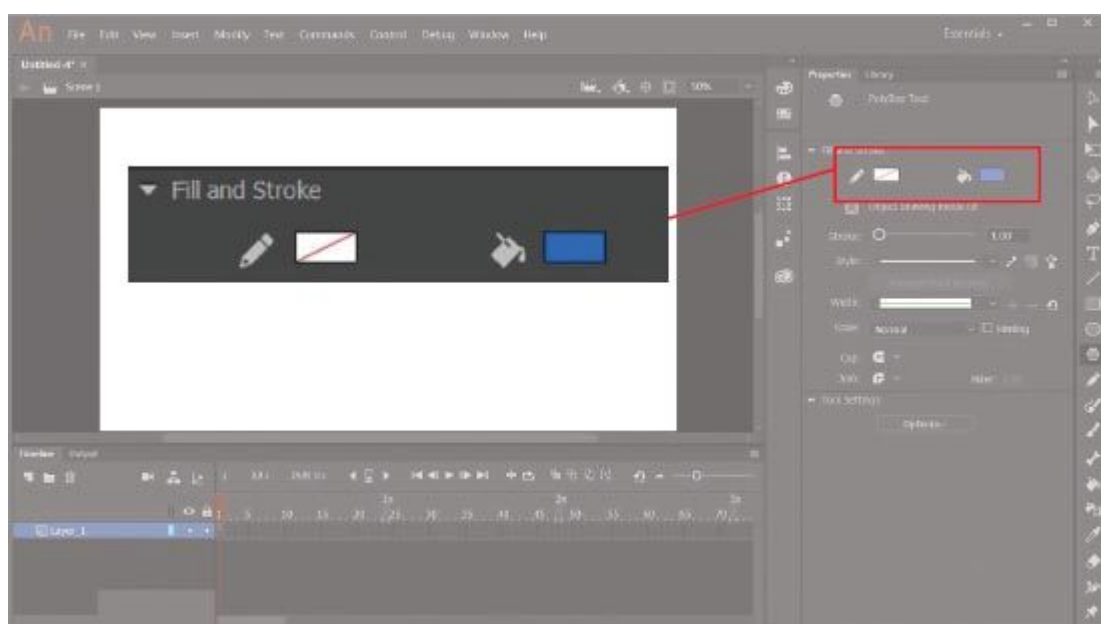
## Controlling object properties

The side toolbar has a range of shapes you can use to create vector graphics. Clicking one of the shapes brings up its **Properties** menu, where you can select various options, such as fill colours, line colours, size and style, before drawing the shape or line on screen (see Figure 20.18).

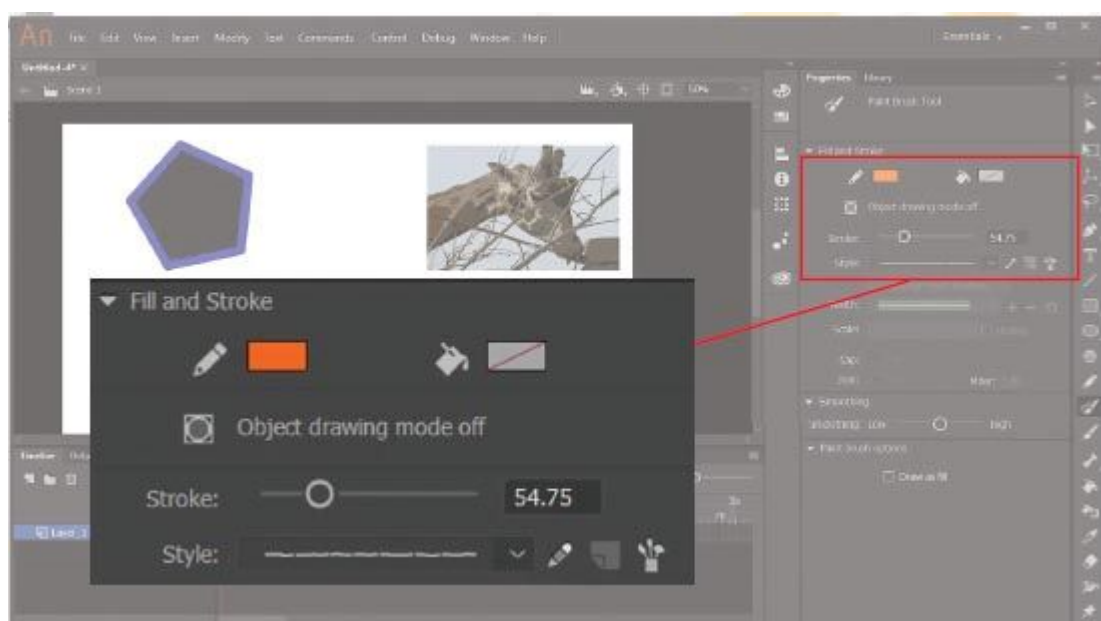
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### Fill and stroke settings

The **Line** and **Fill** options are accessed by clicking on them independently (see Figure 20.19).



**Figure 20.19:** Line and fill options.



**Figure 20.20:** Fill and stroke options.

The menu to choose the colour will then appear. Select the colour from the block.

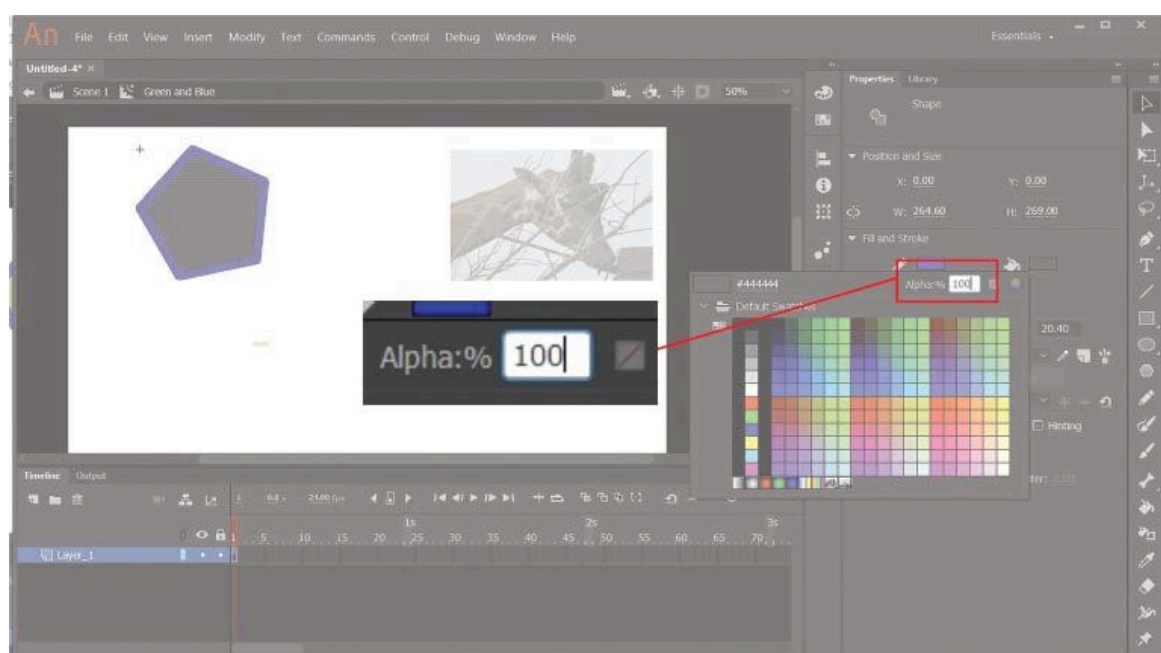
### TIP

Clicking on the sphere will show you a wider range of colours.

When an image is selected using a drawing tool, there are additional properties to choose from.

For example, for the paint brush tool, the **Stroke** option is the size of the line you will draw and there are a range of styles that produce different patterns (see Figure 20.20). There is only a line colour option as the paint brush tool only draws lines; it does not draw shapes to fill.

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**Figure 20.21:** Opacity (Alpha %).

## Transparency

The transparency of an object or a piece of text can be changed. In graphics software, this may be referred to as the opacity or the alpha channel. This is measured as a percentage: 0% opacity will lead to a fully transparent, invisible image, while at 100%, the image is fully visible with no transparency.

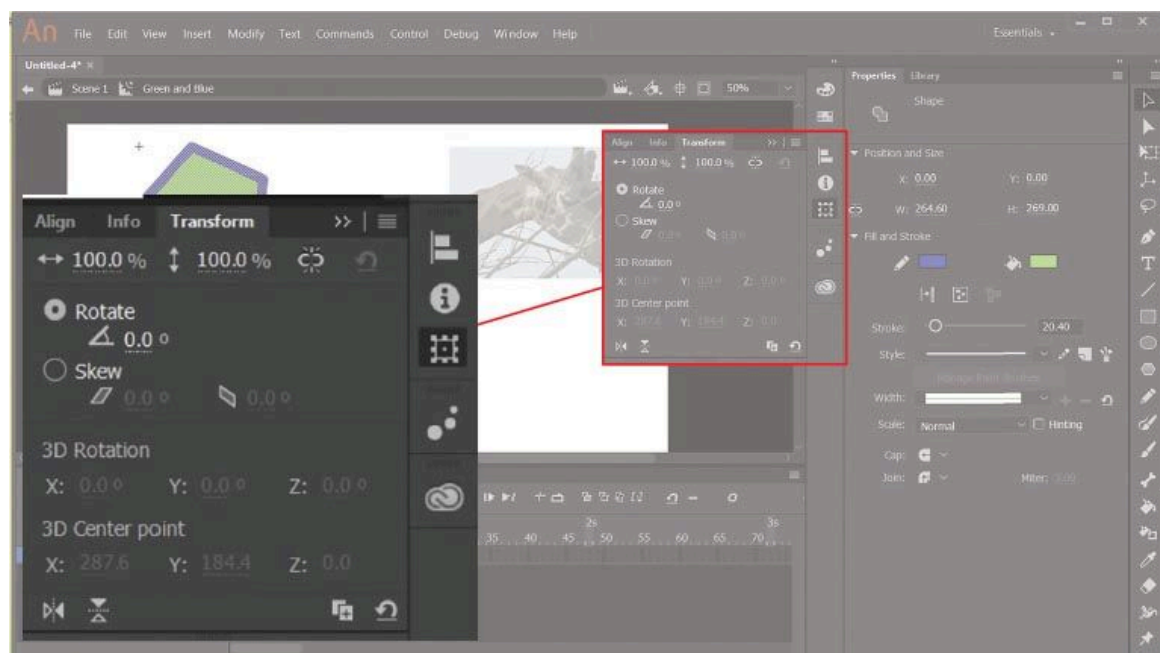
The visibility of an object can be set when it is created. It can also be manipulated during an animation to make objects appear and disappear. To change the opacity of an image, select its **Properties** and go into **Fill** options, as shown in Figure 20.21.

## Size position and orientation (transforming)

To change the appearance of an image, it can be **transformed**. Transform allows the size of the object to be changed, rotated, or have its centre point changed (so it can rotate from a different place on the object). Use the selection tool to select the object to change (hold

down **Shift** and click to select more than one object). Right click on the item, select **Transform** and then **Free Transform**. These options can also be accessed independently from the **Transform** menu, as shown in Figure 20.22.

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**Figure 20.22:** Transform options.

## Questions

- 12 Describe how vector graphics are made.
- 13 What is the difference between line and fill colours?
- 14 Which option do you select to rotate a shape or object?

## Setting paths

The route a symbol takes when moving around the stage is known as a **path**. Defining a path enables frames to be automatically generated between start and end points, which is discussed in more detail when examining inbetweening (‘tweening’) tools next.

## Using layers

An animation is made up of **layers**. Each layer has its own set of frames that can be animated independently. Without layers, the entire image would need to be manipulated within each frame, rather than separate parts.

For example, if a stick person is to be animated to look like they are walking, the elements that need to be separate might be:

- left arm
- right arm
- left leg
- right leg

- head
- body.

These elements might need to be separated further. Depending on the level of detail required, 'left arm' may need to be 'upper left arm' and 'lower left arm'.

Each layer is shown on the **timeline** and needs to be given an appropriate name to describe its function.

To create a new layer, click **Insert** and then **Timeline**, then **Layer**. To change its name, double click the layer name beside the timeline and change the text.

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### TIP

Names cannot contain any spaces.

Folders can also be created for layers. Complex animations may have hundreds of layers, so related layers can be grouped together. For example, consider the stick person walking animation. A folder called 'stick\_person' could be created, with all the layers for that person in the folder. A separate folder could then be created for other layers, such as those for the background.

To create a folder, click **Insert**, then **Timeline**, select **Layer Folder**. Layers can be dragged into the folder and double clicked to rename.

The timeline is made up of frames in chronological order from left to right, as shown in Figure 20.23. Each small rectangle represents a frame, with a filled grey rectangle showing how many frames where the symbol will be displayed.

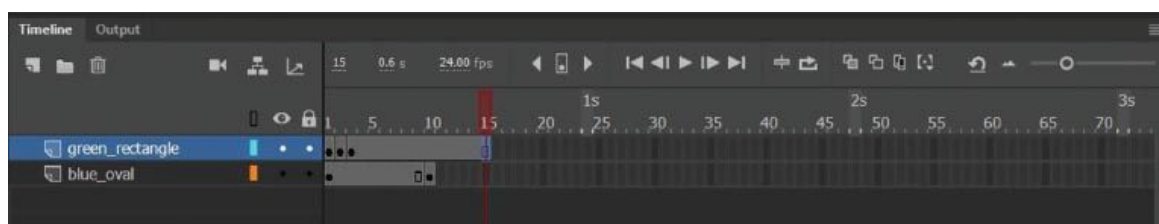
If there is no rectangle for a layer in a specific frame (for example, frame 30), then the items in that frame do not exist when that frame is displayed. In Figure 20.24, 'blue\_oval' does not have a frame beyond frame 10, whereas 'green\_rectangle' has frames up to frame 15. This means on the current frame selected (frame 15 as shown by the red rectangle on the number 15), only the 'green\_rectangle' is visible.

It is important to also order symbols using layers. The symbol in the highest layer in the timeline is in front of all the others.

In Figure 20.25, the layer 'blue\_oval' is at the top, and on the stage it appears in front of the green rectangle.

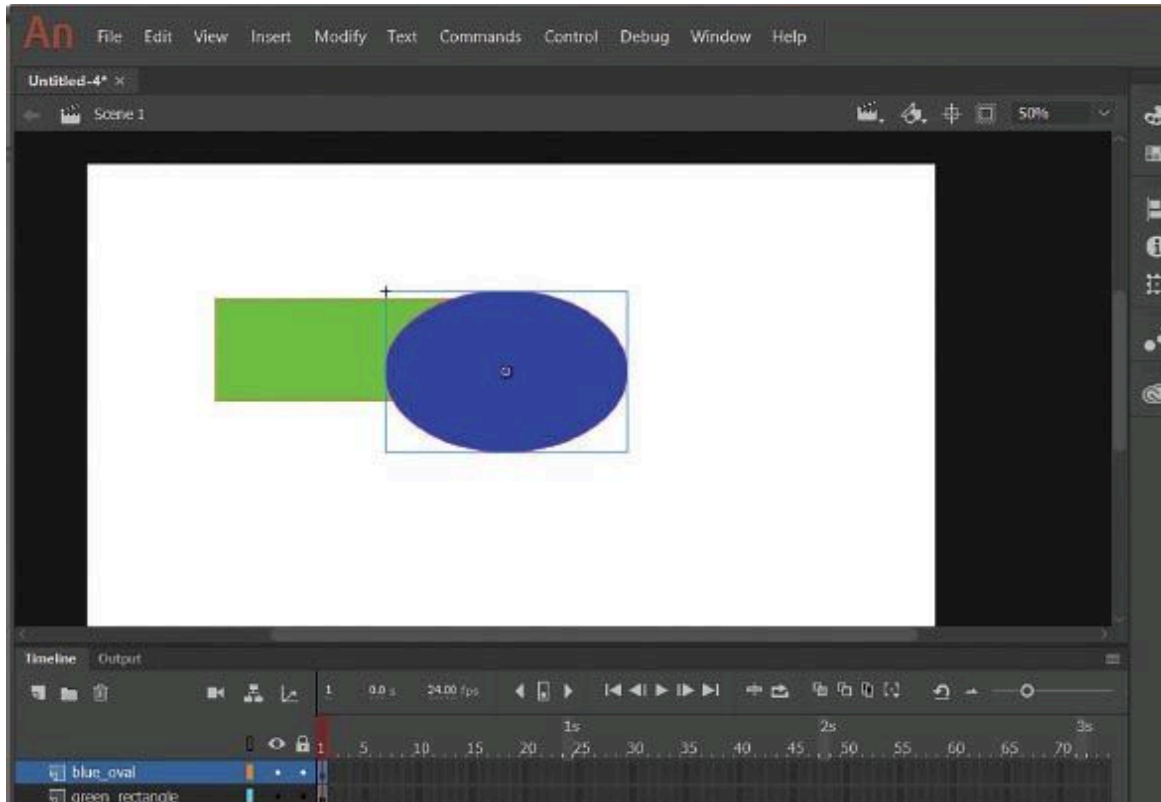


**Figure 20.23:** Timelines.

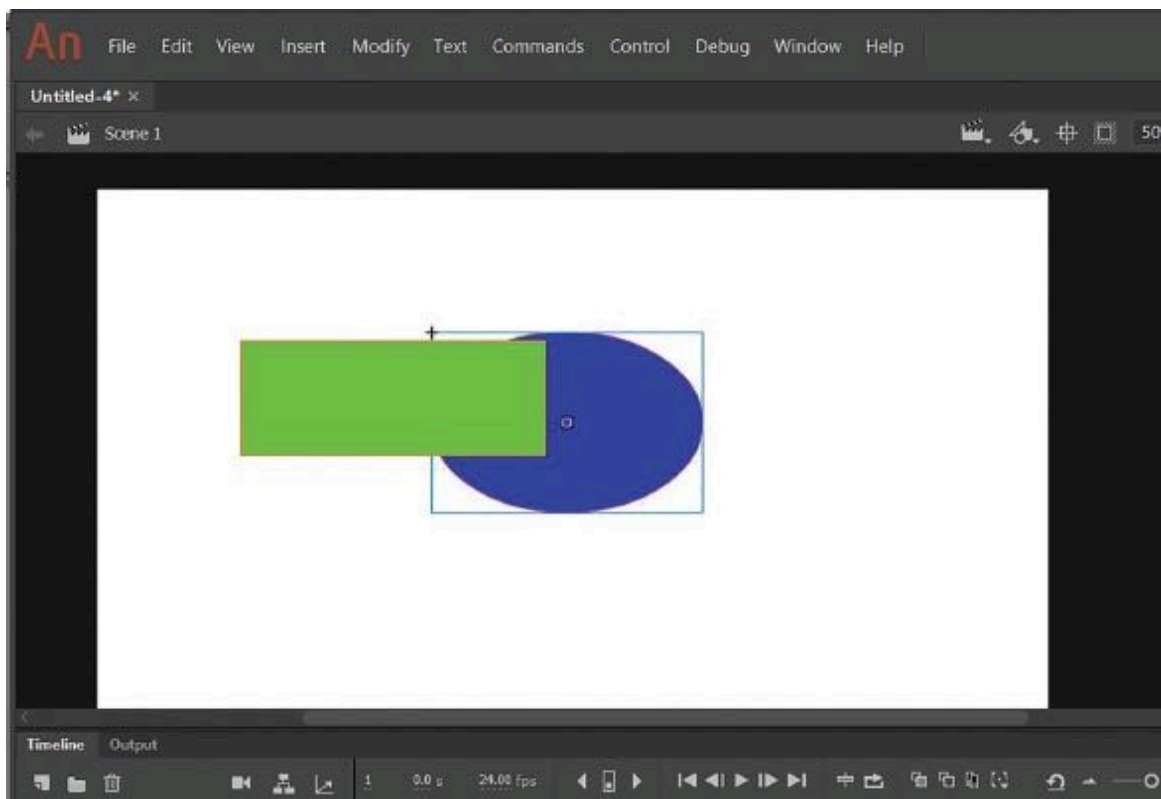


**Figure 20.24:** Green rectangle timeline.

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**Figure 20.25:** Blue oval in own layer overlapping the rectangle.

If the layers are swapped in the timeline, their order will be reversed on the stage. The green rectangle is now in front, as shown in Figure 20.26.

**Figure 20.26:** Order of layers swapped so that there is no deletion of part of the rectangle.



## Questions

15 What is a layer?

16 Why are layers used in animation?

### ACTIVITY 20.03

- 1 Create a stick person, with each part of the stick person in its own layer. Put the stick person layers into a folder. Save the file as **stickPerson** to use later.
- 2 Draw an underwater scene with rocks, seaweed, plants and fish. Make sure each component part has its own layer. Save the program as **underwaterScene** to use later.
- 3 Write your name, with each letter in its own layer. Save the file as **myName** to use later.

## Inbetweening ('tweening')

An animation created solely with frames and key frames can be quite robotic unless very small changes are made between each key frame. Making changes like this can be time consuming. By inserting an 'inbetween' animation (also referred to as simply a **tween**), the computer can generate the animation automatically by filling in the spaces in between key frames. For example, if moving a drawing of a fish across the screen, the start location (A) and end location (B) can be set as key frames, with the computer determining the required movement to get the fish from A to B.

Key frames are represented by a dot in the frame on the timeline, as shown in Figure 20.27.

For example, 'green\_rectangle' is in one position in frame 1. It then changes in frame 2, and frame 3; it then does not change again. 'blue\_oval' is created in frame 1; it does not change then until frame 10.

To add frames and key frames, right click the frame to add a frame or key frame, then choose **Insert Frame** (or **Insert Keyframe**) from the menu.

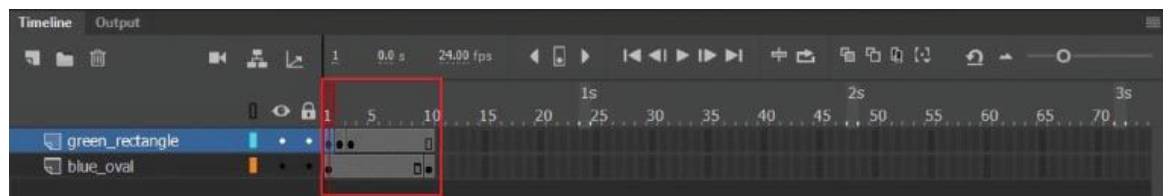
When applying a tween to a layer, it will affect every item in that layer, so it is important to make sure that each individual item has its own layer.

There are two types of tween: motion and shape (including size and colour).

### Motion tween

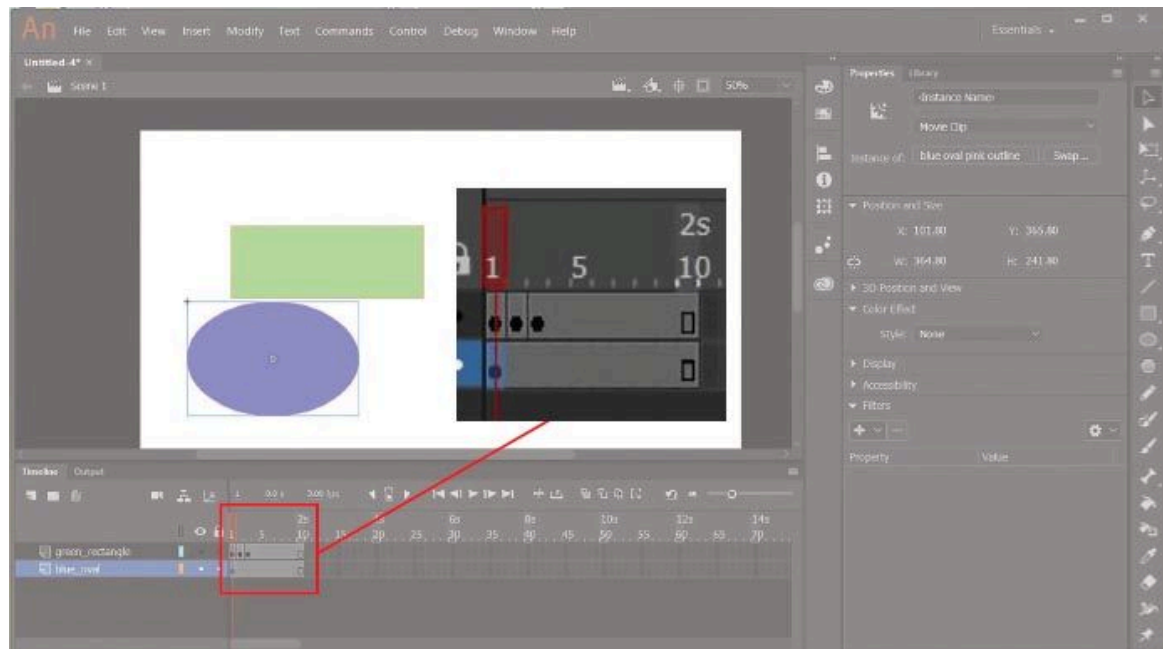
A motion tween only deals with the movement of an object, such as moving from one place to another, and its rotation.

To create a motion tween in Adobe Animate, the item you tween needs to be set as a symbol before starting to move it. Create the first key frame with the symbol in its starting position and add frames up to the point where the tween should end (see Figure 20.28).



**Figure 20.27:** Key frame dots.

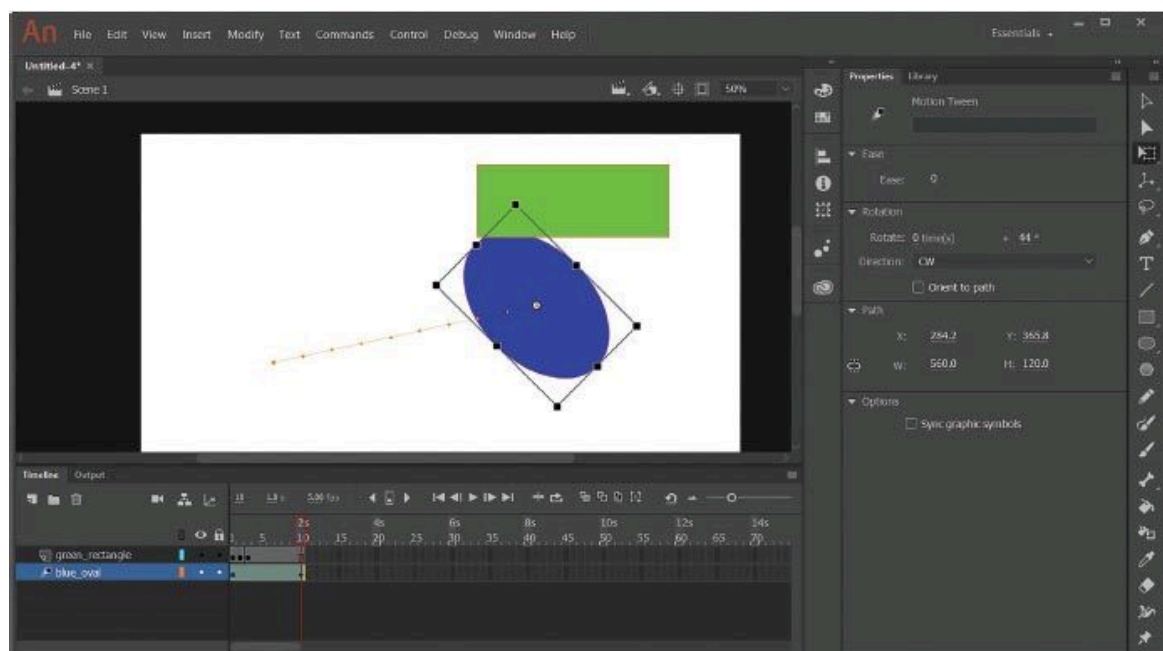
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**Figure 20.28:** Starting a motion tween.

Then, right click the added frames and select **Create Motion Tween**.

Insert a key frame in the frame where the animation should end and then, with this frame selected, move the object to the end position (see Figure 20.29). The object can also be rotated if required.



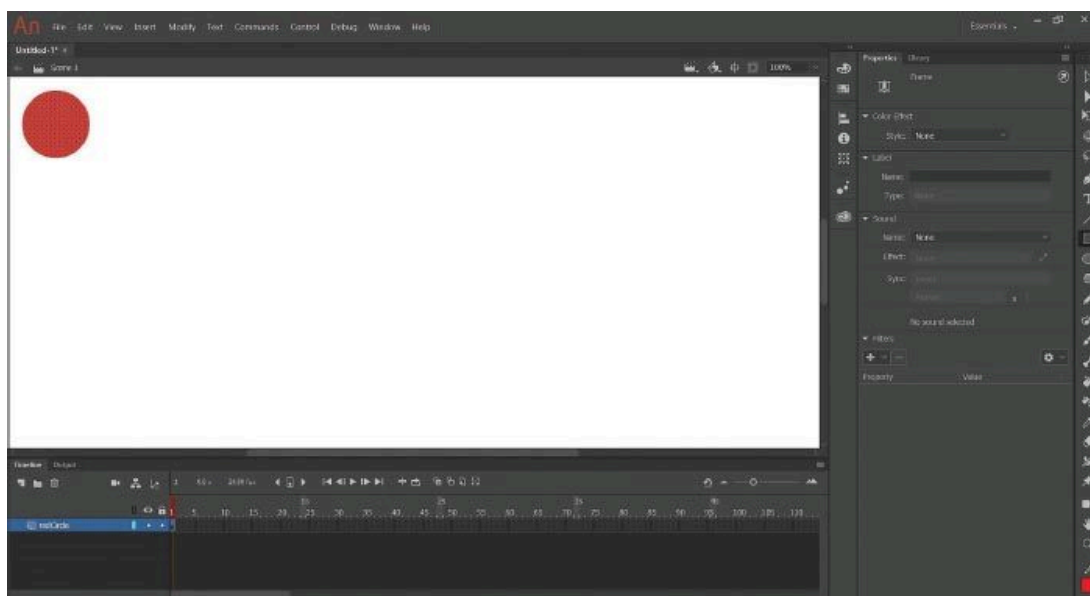
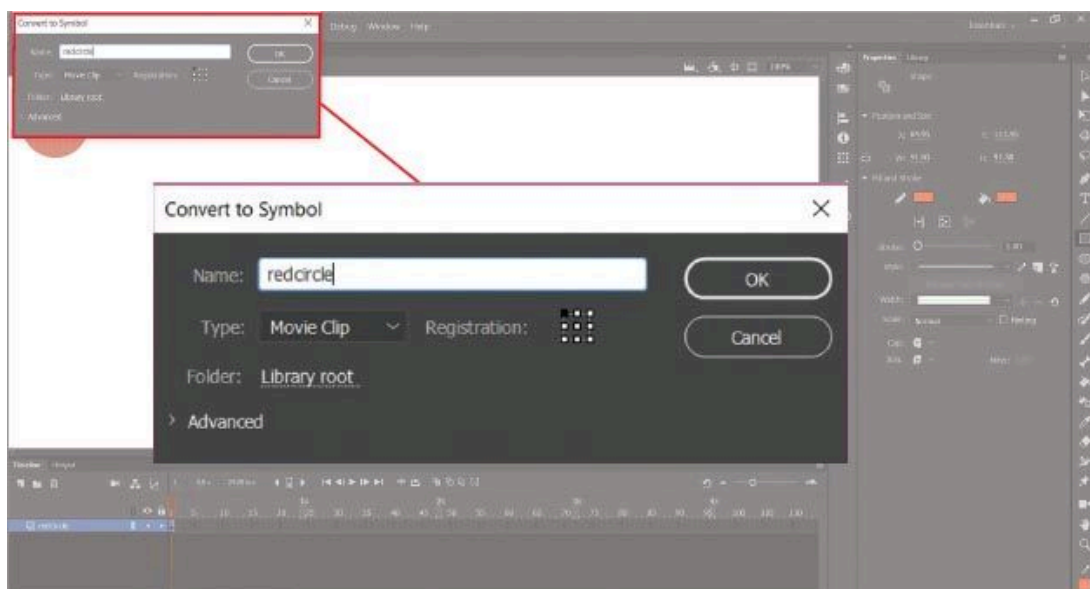
**Figure 20.29:** End of motion tween.

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**WORKED EXAMPLE 20.01**

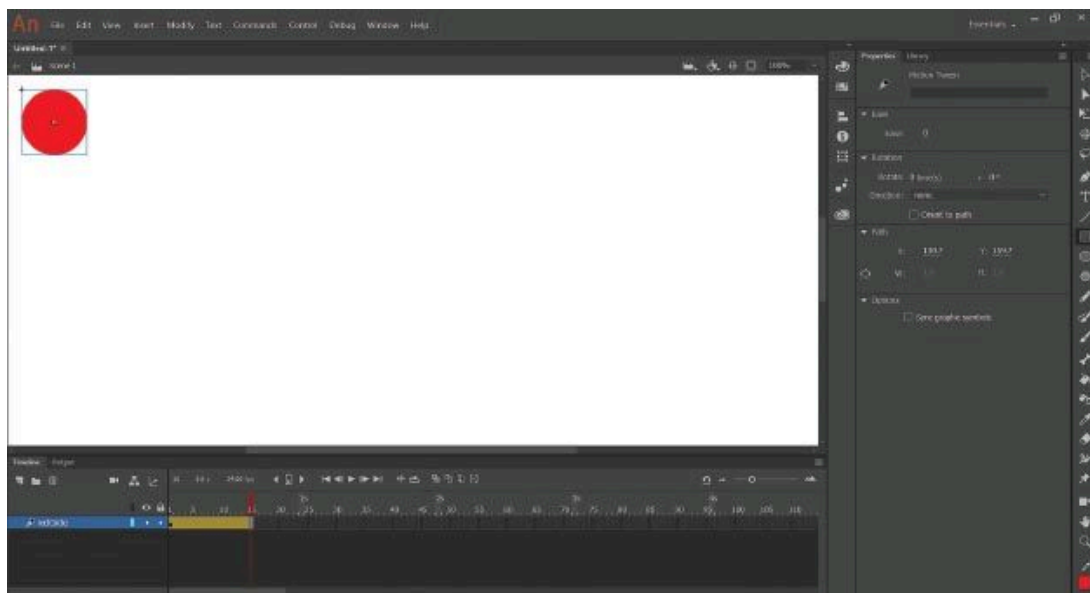
In this example, we will create an animation where a red circle moves to each corner of the screen using inbetweening.

Rename the layer 'redcircle', choose the line colour red and fill colour red, then draw the circle, as shown in Figure 20.30.

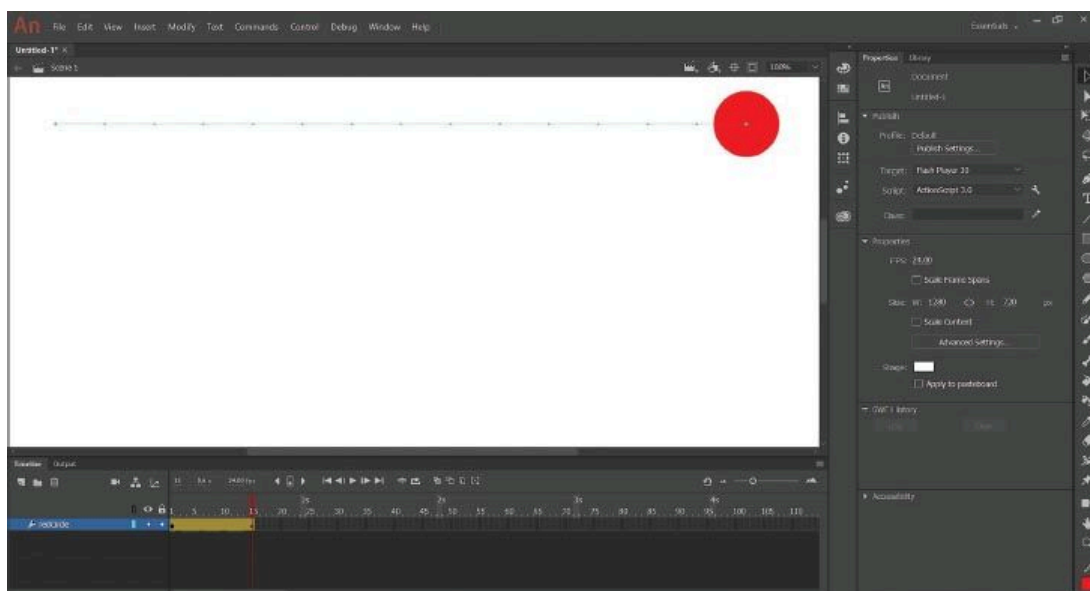
**Figure 20.30:** Rename the layer**Figure 20.31:** Convert the red circle to a symbol.

Convert the red circle to a symbol with a sensible name (see Figure 20.31).

Add frames to frame 15, and then **Create Motion Tween** (see Figure 20.32).



**Figure 20.32:** Add frames.



**Figure 20.33:** Add a key frame.

Add a key frame on frame 15 and move the circle to the top right corner of the screen, as shown in Figure 20.33.

Repeat to make the ball move to all four corners of the stage.

## ACTIVITY 20.04

- 1 Create a ball and make it move around the screen.
- 2 Create a car animation where the car appears on screen and drives around a track that is drawn on the background using a motion tween. Save the file as **carAnimation**.
- 3 Reopen the file **myName**. Use a motion tween to create an animation of each letter of your name flying onto the screen one by one, eventually displaying your full name.
- 4 Add fish to the file **underwaterScene** that you created earlier. Make each one move using a motion tween. Add a bubble for each fish and use a motion tween

to make the bubbles float to the surface.

- 5 Reopen the file **stickPerson**. Create an animation with your stick person. Make them walk across the screen. Challenge yourself and make them perform other actions such as jumping, running and dancing. You could introduce a second person by copying your folder of layers and animating them separately.

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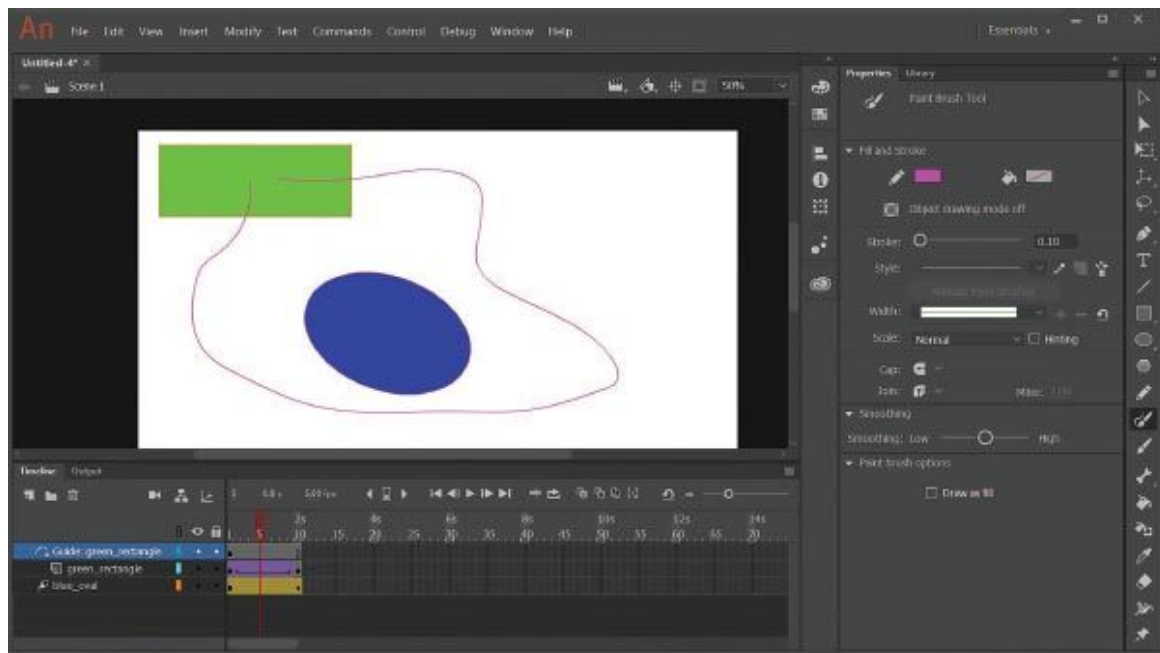
A path can be drawn for a symbol to follow, with a motion tween set to follow this path.

Within Adobe Animate, first add the key frame where the path should start and another where it should end.

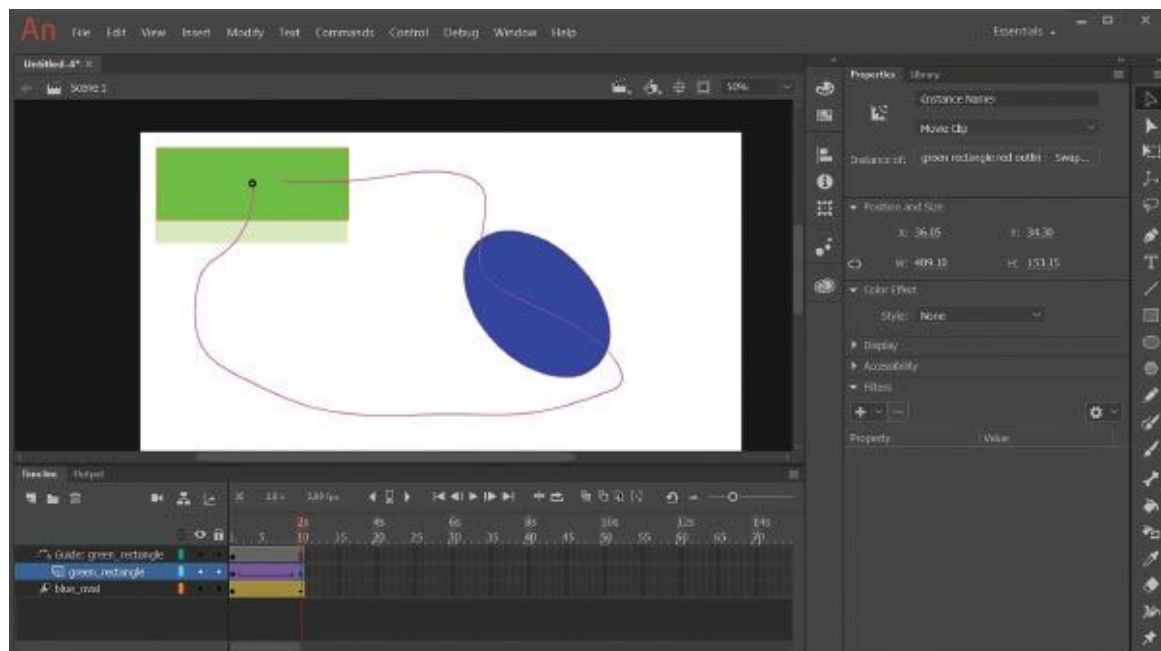
Right click and select **Create Classic Tween**. This will turn the frames blue, along with having an arrow added to the timeline.

Right click the layer and select **Add Classic Motion Guide**, then use the drawing tools to draw the path, as shown in Figure 20.34.

Select the first key frame and drag the object so its centre (the circle that appears) locks onto the start of the path. Then, select the last key frame and drag the object so its centre locks onto the end of the path. That will create the motion path tween (see Figure 20.35).



**Figure 20.34:** Drawing a path.



**Figure 20.35:** Creating a motion path.

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### ACTIVITY 20.05

- 1 Create a red ball. Draw a path for the red ball to follow and animate it so it follows the path.
- 2 Add a second ball with a new path and animate it to follow that path.
- 3 Reopen the file **carAnimation** created earlier. Make the road include a number of bends. Draw a path to make the car move around the track.
- 4 Reopen the file **underwaterScene** created earlier. Add an octopus to the underwater scene. Draw a path for the octopus to follow and animate it.

### Shape tween, including size and colour

A shape tween enables a shape to be changed into a new shape, along with altering the size and colour. For example, a small red square could be turned into a large blue circle, with the computer generating each step of the transformation. A shape tween can also include movement, so the shape will change while it moves.

In Adobe Animate, a shape tween cannot take place on a symbol, as the symbol is in the library and cannot be changed in this way. To create a shape tween, draw an object using the drawing tools, such as a shape or line, but do not convert it to a symbol. Add the object in the key frame where the shape tween should start. Next, add frames to where the shape tween should end, in the same way as a motion tween, as described in the previous section. Right click on the frames on the timeline and select **Create Shape Tween**.

Next, right click in the frame where the tween should finish and then select **Insert key frame**.

With the new key frame selected, either change the colour, shape or size of the image. Alternatively, delete the contents of the frame, and draw a new image.



## Questions

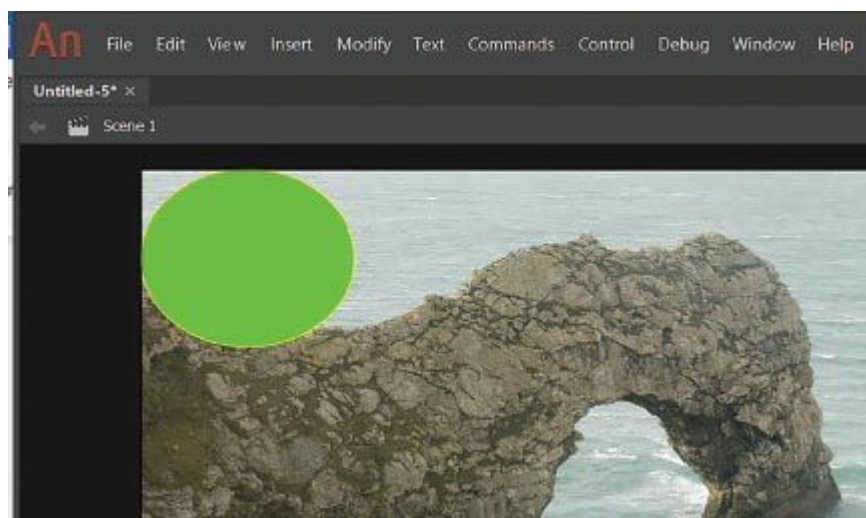
- 17 What is the difference between a motion tween and a shape tween?
- 18 Explain why inbetweening produces smoother animations than frame-by-frame animations.

## Applying masks

A **mask** can make it so that only certain parts of an image are visible. In Adobe Animate, masks are added in their own layers.

To add a mask, right click the layer you want to become a mask and select **Mask**. The layer beneath automatically becomes a sub-layer of the mask.

Draw the shape for the mask in the first layer (see Figure 20.36).



**Figure 20.36:** Mask shape.

Click the **padlock** on the timeline to lock this layer. The image can then be viewed through the masked area.

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Rather than a mask revealing part of an image, it is also possible to add a concealing mask to hide parts of an image. To use a concealing mask, select the required part of the image, then hold Alt (Windows) or Option (Mac) and click Add Layer Mask from the bottom of the Layers section.

A mask can be adjusted by double clicking on the mask in the Channels panel, then changing the opacity and the edges.

Masks can be animated in the same way as an ordinary symbol by removing the padlock to unlock the layer, and then adding the required movement.

## Questions

- 19 Describe the purpose of a mask and give an example of how it can be used.

## Controlling animations

## Adjusting frame rates

An animation's **frame rate** relates to the number of consecutive (following one after the other) frames that are shown each second, which is measured in frames per second (fps). For example, 10 fps means that ten frames will be displayed for each second of the animation, so a 200-frame animation would be 20 seconds in length ( $200 \text{ frames} / 10 \text{ fps} = 20 \text{ seconds}$ ). The higher the frame rate, the quicker the frames change and the smoother the animation appears, although more frames need to be produced.

The frame rate can be changed by entering a different number of frames above the timeline.

### DID YOU KNOW?

Traditional cel animations are often shown at 24 fps, which gives the impression of objects moving smoothly to the viewer. In video games, developers often try to animate objects at 60 fps to make the movement within the game as realistic as possible (see Figure 20.37).



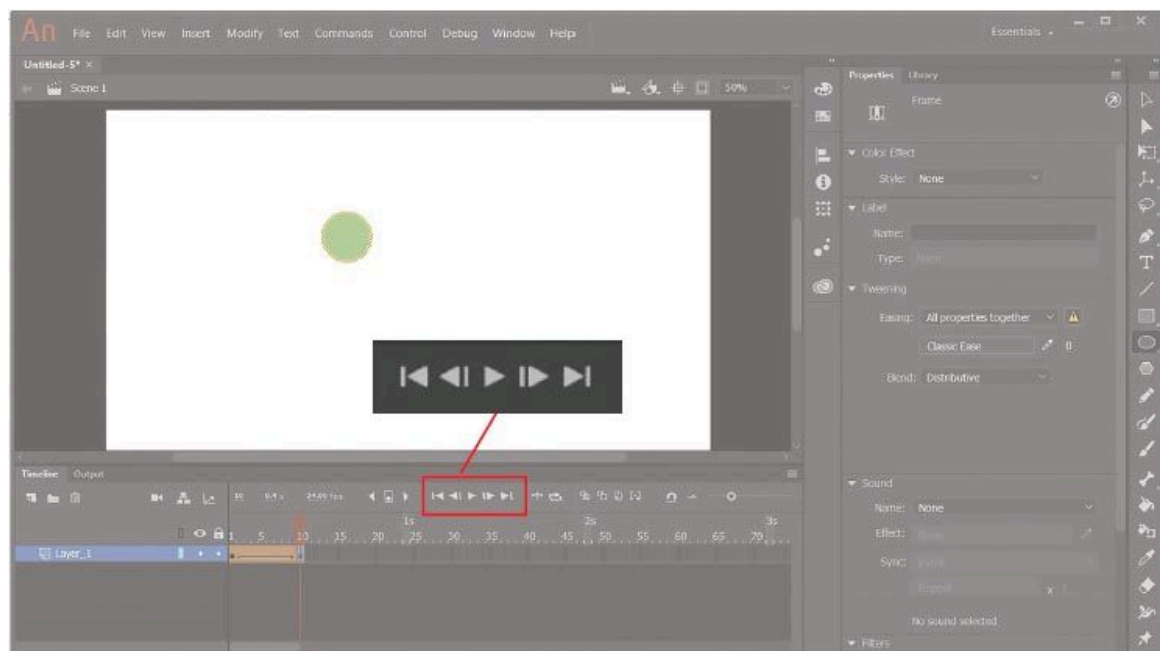
**Figure 20.37:** Video gamer.

## Questions

**20** What effect will increasing the frame rate have on an animation?

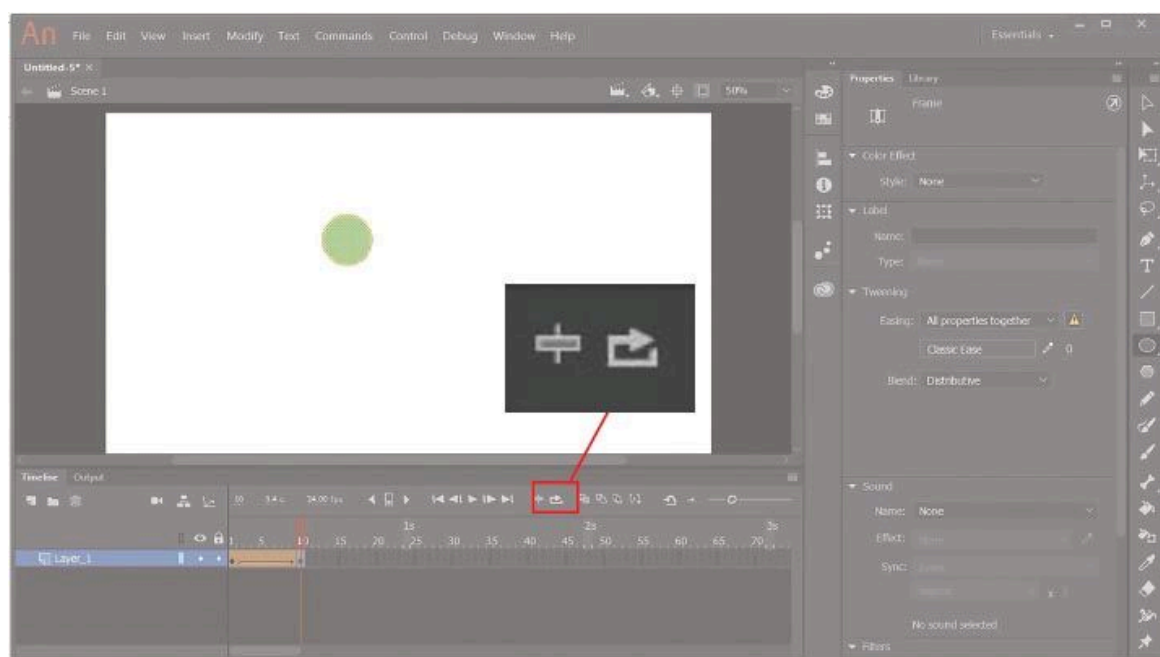
## Looping or stopping animations

To test-run an animation, click **Enter** on the keyboard, or click the play symbol above the timeline, as shown on Figure 20.38. The stepping tool within the animation play controls moves through the animation one frame at a time.

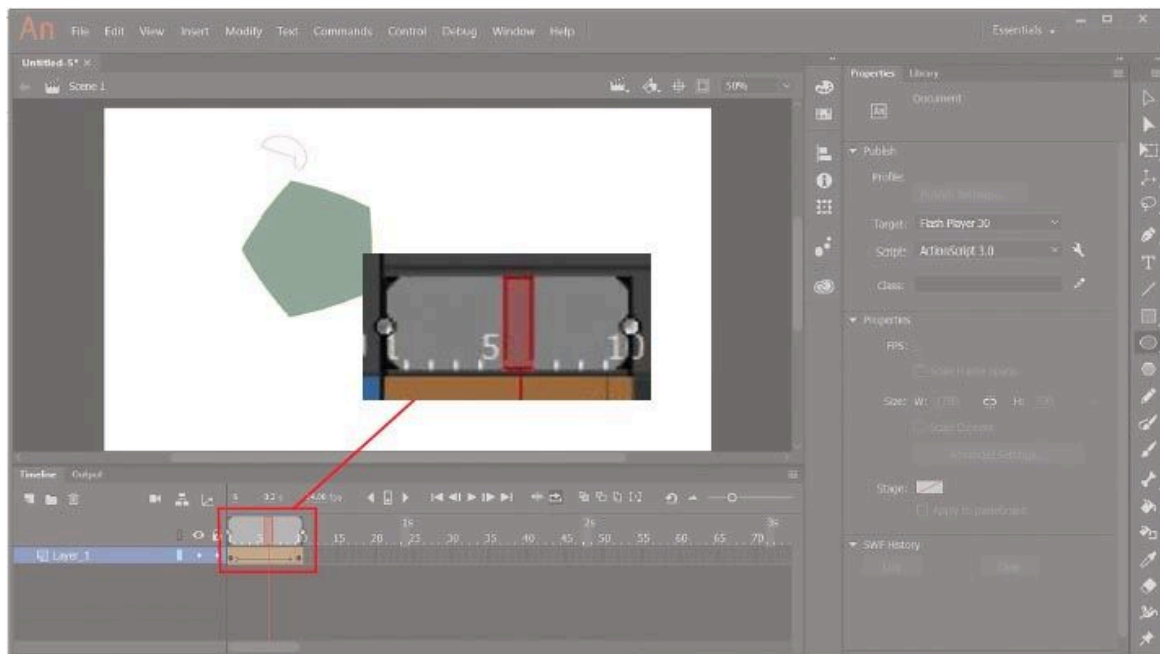


**Figure 20.38:** Animation play symbols.

To watch the video on loop, select the **Loop** option above the timeline, as shown in Figure 20.39.



**Figure 20.39:** Animation loop option.



**Figure 20.40:** Selection of frames to be looped.

Specific frames can be looped by dragging the grey box over the frames (see Figure 20.40).

To stop the animation when it is playing, click the **Play** button. While the animation is playing, the button changes to the pause symbol.

To view the animation looping continually in a separate window, hold down **Control** and press **Enter**.